Virtual Confabulation: A Construct Proposal on the Integration of Immersive Episodic Memories into the Autobiographical Record

André Cicivizzo

Undergraduate Student in Psychology

Universidade Paulista (UNIP)

São Paulo, SP, Brazil

E-mail: andre.cicivizzo@outlook.com

ORCID: https://orcid.org/0009-0008-8414-8403

Abstract

The popularization of immersive virtual reality (VR) devices has raised a question that remains insufficiently theorized in cognitive psychology: the mnemonic fate of simulated experiences encoded with phenomenological features similar to those of physical events. This article proposes the construct of virtual confabulation (also termed immersive autobiographical confabulation) to designate the normative, nonpathological process by which immersive virtual memories, that is, episodic records formed through embodied experience in simulated environments, are retrieved and integrated into the autobiographical record of the physical world, with involuntary distortions of source, content, or temporal context. Grounded in the reconstructive model of episodic memory, the source monitoring framework, and empirical studies on confusion between VR and real-world memories, the proposal distinguishes virtual confabulation from clinical confabulation, generic false memory, and isolated source confusion. It is argued that technological immersion reduces the effectiveness of mechanisms that discriminate simulated from real experience, favoring the incorporation of immersive virtual memories into self-narrative. Implications for clinical psychology, education, forensic contexts, and technology ethics are discussed, along with methodological guidelines for the empirical operationalization of the construct.

Keywords: virtual confabulation; immersive virtual memory; virtual reality; episodic memory; source monitoring; false memories; autobiographical memory

1. Introduction

Autobiographical memory occupies a central place in the organization of personal identity. Far from functioning as a faithful record of past events, it operates as a reconstructive system, susceptible to omissions, distortions, and narrative integrations that aim to preserve the subjective coherence of the self (Conway & Pleydell-Pearce, 2000; Schacter & Addis, 2007). This plasticity, under ordinary conditions, does not constitute pathology. Rather, it reflects the adaptability of a system evolutionarily oriented toward present and future guidance rather than exact historical preservation (Schacter, 1999).

Over recent decades, cognitive psychology has extensively investigated false memories produced by verbal suggestion, repetition, vivid imagination, and confusion among distinct sources, such as photographs, third-party accounts, and films (Loftus, 2005; Johnson, Hashtroudi, & Lindsay, 1993). In general, these sources maintain some phenomenological distance from direct experience: the subject tends to know, even if confusedly, that they saw an image, heard a story, or imagined a scene. Virtual reality alters this condition.

By occupying the visual field, incorporating bodily movement, simulating perceptual consequences, and mobilizing emotional and autonomic responses, VR approximates the experiential structure of physical events in a way that has no precedent among media technologies available to the general public (Slater & Wilbur, 1997; Smith, 2019). Recent studies indicate that participants confuse memories originating in VR with memories of physical events with statistically relevant frequency (Brade et al., 2021; Brade et al., 2024). Hoffman et al. (2001) had already described virtual reality monitoring, the process of discriminating between real and virtually encoded memories during retrieval. However, the literature still lacks a construct that integrates, within a single theoretical framework, the formation of immersive simulated memories, their possible autobiographical integration, and the involuntary distortions associated with this process.

We propose, in this article, the term virtual confabulation to name this phenomenon. The construct designates the mnemonic displacement by which immersive virtual memories cease to be retrieved as records of simulation and begin to function, wholly or partially, as memories of events that occurred in the physical world. The proposal does not intend to replace consolidated categories, but to organize them in the face of a new technological condition of experiential encoding.

This text has four objectives: (a) to provide a theoretical foundation for normative virtual confabulation; (b) to operationally define the constructs immersive virtual memory and virtual confabulation; (c) to describe their mechanisms and forms of manifestation; (d) to indicate practical implications and guidelines for empirical investigation.

2. Theoretical Foundation

2.1. Episodic Memory and Autobiographical Reconstruction

Episodic memory refers to the capacity to recall personally experienced events situated in time and context (Tulving, 2002). Its functioning does not correspond to simple reactivation of mnemonic traces, but to a reconstructive process guided by schemas, expectations, and affective states present at the moment of retrieval (Bartlett, 1932; Schacter & Addis, 2007). Plausible yet nonexistent details may be incorporated into autobiographical accounts without deliberate intent to falsify. Memory operates, to a large extent, by narrative coherence rather than documentary fidelity.

Experiences in VR enter this reconstructive system bearing characteristics typically associated with lived events: egocentric perspective, motor interaction, sensory feedback, and emotional modulation. When retrieved, they tend to be treated not as externally received information, but as raw material of personal past. This condition favors, in principle, the integration of immersive virtual memories into autobiography, especially when the subject lacks, at the moment of retrieval, reliable cues about the simulated origin of the experience.

2.2. Source Monitoring and Confusion Between VR and Reality

According to the source monitoring framework (Johnson, Hashtroudi, & Lindsay, 1993), identification of the origin of a memory depends on characteristics encoded at the moment of engram formation, such as perceptual richness, contextual information, cognitive effort, emotional coherence, and congruence with prior knowledge. Failures in this process produce source confusion, when content is remembered but incorrectly attributed to direct perception, imagination, or another origin.

VR complicates this monitoring because it simulates precisely the experiential properties that the mnemonic system uses to identify veridical perceptions. Brade et al. (2021) demonstrated that participants confuse VR and reality memories more frequently than they confuse monitor presentation and reality memories. Brade et al. (2024), in a laboratory study, confirmed the occurrence of source confusion involving VR and identified fragile heuristics used by participants to discriminate memories, such as assumptions about graphical limitations of the technology. Such heuristics lose effectiveness as immersive devices become more realistic.

Hoffman et al. (2001) showed that discrimination between real and virtually encoded memories is based, in part, on differences in phenomenal characteristics associated with each source, in accordance with the model of Johnson and Raye (1981). However, the authors also observed that participants attributed similar characteristics to false memories of new items, suggesting that part of monitoring involves posterior reconstructive inference rather than mere reading of originally encoded properties.

2.3. Clinical Confabulation and the Proposed Extension

In neuropsychology, confabulation designates the spontaneous or provoked production of false or distorted memories, without intent to deceive and frequently accompanied by subjective conviction of veracity (Kopelman, 1987; Schnider, 2008). Classically associated with frontal, diencephalic, or amnesic conditions, clinical confabulation reflects disruptions in executive control over retrieval and verification of autobiographical contents.

Virtual confabulation, as proposed here, does not presuppose neurological injury. It is a normative mnemonic distortion, mediated by immersive technology, observable in cognitively intact individuals. The terminological parallel is justified by phenomenological similarity: in both cases, the subject retrieves incorrect autobiographical contents without awareness of error and without deceptive purpose. The difference lies in etiology. In clinical confabulation, organic dysfunction predominates; in virtual confabulation, a technological condition of experiential encoding predominates.

It is important to emphasize that the proposal does not medicalize VR use nor pathologize healthy users. The qualifier normative is therefore essential: virtual confabulation names a predictable cognitive outcome of reconstructive memory exposed to convincing simulation, not a psychiatric or neurological syndrome.

3. Operational Definition of the Constructs

3.1. Immersive Virtual Memory

We define immersive virtual memory as the episodic record of an experience encoded exclusively through immersive interaction in a virtual reality environment, encompassing perceptual, spatial, motor, emotional, and interactional contents of simulated origin. This term is preferred over virtual memory alone because, in computer science, the latter commonly refers to memory management in operating systems. Immersive virtual memory should not be confused with memory of information about VR (for example, remembering having read an article about headsets), nor with memory of content passively watched on a two-dimensional screen. It presupposes immersive experience, even if partial, mediated by a VR device.

For brevity, and when context excludes ambiguity, we use immersive virtual memory throughout this article.

3.2. Virtual Confabulation

We define virtual confabulation (also termed immersive autobiographical confabulation) as the normative process, and in a derivative sense the product of this process, by which immersive virtual memories are retrieved, reconstructed, and integrated into the autobiographical record of the physical world, producing one or more of the following involuntary distortions:

(a) incorrect source attribution, when virtually originated content is remembered as a physical event;

(b) hybrid fusion, when virtual and real elements combine into a memory of an event that did not occur in either modality in isolation;

(c) confabulatory filling-in, when mnemonic gaps are completed with material inferred or recombined from simulated fragments.

In all cases, virtual confabulation involves subjective conviction of veracity and absence of intent to deceive. Its distinctive criterion is not merely error of origin, but the functionalization of immersive virtual memory as an autobiographical experience of the physical world.

3.3. Summary of Differential Criteria

For conceptual clarity, we propose the following minimum criteria for identification of virtual confabulation in investigative contexts:

1. original encoding of the experience in a VR environment through immersive interaction;
2. posterior retrieval with total or partial attribution to the physical world;
3. absence of deliberate intent to falsify;
4. reported or inferred subjective conviction from the account;
5. verifiable distortion by comparison with a record of the virtual session, experimental protocol, or other reliable external source;
6. absence of neurological injury or clinical amnesic syndrome sufficient to explain the distortion.

4. Mechanisms of Virtual Confabulation

4.1. Immersive Encoding

During use of VR headsets, visual, auditory, proprioceptive, and emotional stimuli are processed in an integrated manner. Neuroimaging evidence indicates partial overlap between networks activated by real experiences and by immersive VR tasks, especially in regions associated with episodic memory and spatial navigation, such as the hippocampus and parietal cortex (Smith, 2019). The greater the depth of semantic, emotional, and motor processing, the greater the tendency toward mnemonic consolidation (Craik & Lockhart, 1972).

Emotionally intense experiences in VR tend to generate more persistent and intrusive immersive virtual memories, in analogy to flashbulb memories of real events (Kensinger, 2009). Emotionally salient contents receive priority in autobiographical reconstruction, which increases the probability of posterior virtual confabulation.

4.2. Storage Without Stable Origin Labeling

Episodic memory does not reliably preserve the categorical distinction between "real" and "virtual." What persists are, above all, traces of how the event was experienced: perspective, movement, arousal, spatial layout, interactions performed. When simulation reproduces such characteristics, the immersive virtual engram may become, in storage, phenomenologically close to engrams of physical events.

4.3. Reconstructive Retrieval and Narrative Integration

Remembering implies recomposing the past in light of current schemas. Immersive virtual memories may be assimilated into preexisting narratives ("I have always been afraid of heights," "I already knew that type of environment"), facilitating their incorporation as real experiences. The search for autobiographical coherence described by Conway and Pleydell-Pearce (2000) favors integration of virtual material compatible with the subject's self-image.

4.4. Failure in Source Monitoring

The critical moment of virtual confabulation occurs at retrieval, when the source monitoring system fails to adequately identify the simulated origin of the engram. This may result from decay of distinctive encoding cues, high phenomenological similarity between VR and reality, or absence of metacognitive knowledge about the virtual experience. Prolonged or repeated sessions tend to dilute the boundary between worlds, especially when the subject does not perform mnemonic delimitation procedures after device use.

5. Forms of Manifestation

Virtual confabulation does not constitute a homogeneous category. We propose three main forms, derived from patterns described in the literature on source confusion and false memories, respecified for the immersive context.

Incorrect source attribution. The subject recalls, with relative accuracy, the content of the virtual experience, but attributes it to the physical world. Example: claiming to have visited a building whose prototype existed only in simulation. This corresponds, in classical terms, to source confusion between VR and reality (Brade et al., 2021).

Hybrid fusion. Elements of immersive virtual memories combine with elements of real memories, producing autobiographical events that never occurred in either modality in isolation. Example: remembering a fall on the stairs at home when the fall was simulated in VR, but the remembered location corresponds to the actual residence.

Confabulatory filling-in. Gaps in virtual or real memory are completed with content inferred or recombined from simulated fragments, generating coherent yet fictitious narrative sequences. Phenomenologically, this approximates provoked confabulation described in neuropsychology, although without identifiable organic substrate.

6. Distinction From Related Constructs

Construct / Central characteristic / Relation to virtual confabulation

False memory / Recall of nonexistent or distorted event / Broad category; virtual confabulation is a VR-mediated subtype

Source confusion / Error in identifying engram origin / Central mechanism, but does not exhaust the phenomenon

Clinical confabulation / Pathological distortion after injury / Phenomenological parallel; distinct etiology

Media memory / Record of passive content / Lower bodily experience and presence

Virtual reality monitoring / Discrimination between real and virtually encoded memories / Direct empirical antecedent

Immersive virtual memory / Mnemonic product of embodied simulated experience / Prior stage to virtual confabulation

The proposed contribution consists in naming the complete trajectory, from immersive encoding to incorrect autobiographical integration, and not merely the punctual error of source attribution.

7. Theoretical and Practical Implications

7.1. Theoretical Contribution

Virtual confabulation allows dispersed literature on false memories, source monitoring, presence, and memory in VR to be reorganized under a unitary framework. It also updates classics of cognitive psychology in the face of a new technological condition of experiential encoding. The construct suggests that the metaphysical distinction between real and virtual matters less, for memory, than the functional and experiential richness of the encoded event.

7.2. Clinical and Neuropsychological Psychology

VR is used in exposure protocols for phobias, post-traumatic stress disorder, and cognitive rehabilitation (Maples-Keller et al., 2017). Its effectiveness is related, in part, to convincing simulation of lived experiences. This same property implies risk of virtual confabulation: memories of simulated exposures may later be recalled as real traumatic or corrective events, with unpredictable consequences for patient self-narrative. Clinical protocols could include structured debriefing and external recording of the virtual condition, procedures still poorly standardized.

7.3. Education and Professional Training

Immersive simulations in medicine, aviation, military training, and historical education produce immersive virtual memories susceptible to confabulation. Students may recall procedures, environments, or interactions they only simulated as field experience, with implications for competence assessment and distinction between procedural knowledge and autobiographical experience.

7.4. Forensic Context

Increasing use of VR in witness training, crime scene reconstruction, and event simulation raises risks of mnemonic contamination (Loftus, 2005). Simulated details, such as spatial layout, characteristics of individuals, and temporal sequences, may integrate into accounts later collected as spontaneous reports.

7.5. Technology Ethics

Virtual environments controlled by commercial platforms, advertising campaigns, or persuasive experiences may implant pseudo-autobiographical experiences. Virtual confabulation describes a cognitive mechanism by which immersive manipulation ceases to operate only at the level of immediate persuasion and begins to modify long-term mnemonic records.

8. Methodological Proposal for Empirical Investigation

Operationalization of virtual confabulation requires an experimental design that separates encoding, retention, and retrieval, controlling immersion, emotional content, and prior familiarity with VR.

8.1. Suggested Experimental Design

We propose a paradigm with three conditions: immersive experience in VR, 360° video of the same duration and content, and written narrative control. After retention intervals of 24 hours and seven days, free recall, cued recall, and source attribution tests are applied, with alternatives such as "occurred in VR," "occurred in real life," and "I am not sure." Plausible lure items, inspired by the paradigm of Loftus and Pickrell (1995), allow quantification of confabulatory filling-in.

8.2. Variables

Dependent variables: rate of incorrect attribution to reality; number of spontaneous confabulations; hybrid fusion index; phenomenological vividness; subjective conviction.

Independent variables: exposure modality (VR, video, text); presence level (Slater-Usoh-Steed Presence Questionnaire); self-reported emotional intensity.

Moderators: prior experience with VR; susceptibility to false memories; age; familiarity with simulated content.

8.3. Hypotheses

H1: Experiences encoded in VR will produce significantly higher rates of virtual confabulation than less immersive conditions.

H2: Higher levels of presence will mediate the relation between immersive condition and virtual confabulation.

H3: Emotionally intense events in VR will increase hybrid confabulations and confabulatory filling-in.

H4: Immediate structured debriefing will reduce, but not eliminate, virtual confabulation after seven days.

9. Limitations and Research Agenda

The proposed construct remains, at this stage, predominantly theoretical. Although the literature confirms source confusion between VR and reality, longitudinal studies on autobiographical integration of immersive virtual memories remain scarce. It is not known, with precision, how long such memories remain susceptible to confabulation, nor which individual profiles are most vulnerable.

Open questions include: Do children and adolescents, in a critical phase of identity formation, show greater susceptibility? Does virtual confabulation decay spontaneously or stabilize as a durable mnemonic trait? Are there cross-cultural differences in attribution of veracity to simulated experiences? Can specific instruments for assessing virtual confabulation be psychometrically validated?

Future investigations should combine behavioral measures, phenomenological self-report, and, when possible, functional neuroimaging during comparative retrieval of immersive virtual and real memories.

10. Final Considerations

Virtual reality does not produce merely temporary perceptions. It produces candidates for autobiographical memories. When the mnemonic system fails to correctly label the simulated origin of these experiences, immersive virtual memories shift into the record of the self as if they belonged to the physical world. Virtual confabulation, understood as normative immersive autobiographical confabulation, names this process.

It is a phenomenon distinct from clinical confabulation, although phenomenologically convergent, and distinct from generic false memory, although mechanistically related. Its specificity resides in immersive technological mediation and in the production of memories that are born virtual and, upon retrieval, confabulate themselves into the real past.

As VR headsets consolidate as domestic, clinical, and educational instruments, understanding virtual confabulation ceases to be a marginal theoretical exercise and becomes a legitimate agenda for cognitive psychology, neuropsychology, and studies on technology and cognition. The construct offers shared vocabulary for investigating how immersive simulation silently rewrites the autobiography of its users.

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André Cicivizzo: conceptualization, literature review, writing (original draft), writing (review and editing).

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